Chapter 14: Second Derivatives (Đạo hàm Cấp hai)

Chapter Overview

The **second derivative** is the derivative of the first derivative. While the first derivative tells us about the rate of change and slope of a function, the second derivative provides information about how that rate of change is itself changing. This concept is fundamental in understanding **acceleration** in physics, **concavity** in curve sketching, and **optimization** in economics.

In this chapter, we will:

- Define the second derivative and higher-order derivatives
- Learn how to compute second derivatives using derivative rules
- Understand the physical interpretation of second derivatives as acceleration
- Apply second derivatives to analyze the concavity of functions
- Solve real-world problems involving acceleration and motion

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14.1 Bilingual Vocabulary (Từ vựng Song ngữ)

Core Concepts

English	IPA	Vietnamese	Example
Second derivative	/ˈsɛkənd dɪˈrɪvətɪv/	Đạo hàm cấp hai	The second derivative of $f(x) = x^3$ is $f''(x) = 6x$
Higher-order derivative	/ˈhaɪər ˈɔːrdər dɪˈrɪvətɪv/	Đạo hàm cấp cao	The third derivative $f'''(x)$ is a higher-order derivative

Acceleration	/əkˌsɛləˈreɪʃən/	Gia tốc	Acceleration is the second derivative of position
Instantaneous acceleration	/ˌɪnstənˈteɪniəs ək ˌsɛləˈreɪʃən/	Gia tốc tức thời	The instantaneous acceleration at $t = 2$ is $a(2) = 6$ m/s²
Concavity	/kɒnˈkævɪti/	Tính lõm	The concavity of a curve is determined by the second derivative
Concave up	/kɒnˈkeɪv ʌp/	Lõm lên (lồi)	If $f''(x) > 0$, the graph is concave up
Concave down	/kɒnˈkeɪv daʊn/	Lõm xuống (lõm)	If $f''(x) < 0$, the graph is concave down
Inflection point	/ɪnˈflɛkʃən pɔɪnt/	Điểm uốn	An inflection point occurs where $f''(x) = 0$ and concavity changes
Velocity	/vəˈlɒsɪti/	Vận tốc	Velocity is the first derivative of position
Position function	/pəˈzɪʃən ˈfʌŋkʃən/	Hàm vị trí	The position function $s(t)$ describes location over time

Mathematical Notation

English	IPA	Vietnamese	Notation
f double prime	/ɛf ˈdʌbəl praɪm/	f hai phẩy	$f''(x)$
f triple prime	/ɛf ˈtrɪpəl praɪm/	f ba phẩy	$f'''(x)$
d squared y dx squared	/diː skweərd waɪ diː ɛks skweərd/	d bình phương y trên dx bình phương	$\frac{d^2y}{dx^2}$
nth derivative	/ɛnθ dɪˈrɪvətɪv/	Đạo hàm cấp n	$f^{(n)}(x)$

14.2 Definition and Notation (Định nghĩa và Ký hiệu)

The Second Derivative

> **Definition: Second Derivative**
>
> If f is a differentiable function, then the **second derivative** of f, denoted $f''(x)$ or $\frac{d^2y}{dx^2}$, is the derivative of the first derivative:
>
> $$f''(x) = \frac{d}{dx}[f'(x)]$$
>
> Or equivalently:
>
> $$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

Common Notations:

If

$y = f(x)$

, the second derivative can be written as:

- $f''(x)$
  (prime notation)
- $\frac{d^2y}{dx^2}$
  (Leibniz notation)
- $\frac{d^2f}{dx^2}$
  (alternative Leibniz notation)
- y''
  (simplified prime notation)

Higher-Order Derivatives

The process can be continued to find **third, fourth, and higher-order derivatives**:

Order	Prime Notation	Leibniz Notation	Name
First	$f'(x)$	$\frac{dy}{dx}$	First derivative
Second	$f''(x)$	$\frac{d^2y}{dx^2}$	Second derivative
Third	$f'''(x)$	$\frac{d^3y}{dx^3}$	Third derivative
Fourth	$f^{(4)}(x)$	$\frac{d^4y}{dx^4}$	Fourth derivative
nth	$f^{(n)}(x)$	$\frac{d^ny}{dx^n}$	nth derivative

Note: For derivatives of order four and higher, we typically use the notation

$f^{(n)}(x)$

instead of multiple primes.

14.3 Computing Second Derivatives (Tính Đạo hàm Cấp hai)

To find the second derivative, we simply **differentiate the first derivative**.

Example 14.1: Polynomial Function

Find the second derivative of

$$f(x) = 3x^4 - 5x^3 + 2x^2 - 7x + 1$$

Solution:

Step 1: Find the first derivative using the power rule:

$$f'(x) = 12x^3 - 15x^2 + 4x - 7$$

Step 2: Differentiate again to find the second derivative:

$$f''(x) = \frac{d}{dx}[12x^3 - 15x^2 + 4x - 7]$$

$$f''(x) = 36x^2 - 30x + 4$$

Answer:

$$f''(x) = 36x^2 - 30x + 4$$

Example 14.2: Exponential Function

Find the second derivative of

$$y = e^{2x}$$

Solution:

Step 1: Find the first derivative using the chain rule:

$$\frac{dy}{dx} = e^{2x} \cdot 2 = 2e^{2x}$$

Step 2: Differentiate again:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}[2e^{2x}] = 2 \cdot e^{2x} \cdot 2 = 4e^{2x}$$

Answer:
$\frac{d^2y}{dx^2} = 4e^{2x}$

Example 14.3: Trigonometric Function

Find the second derivative of
$f(x) = \sin(3x)$

Solution:

Step 1: Find the first derivative:

$$f'(x) = \cos(3x) \cdot 3 = 3\cos(3x)$$

Step 2: Find the second derivative:

$$f''(x) = 3 \cdot [-\sin(3x)] \cdot 3 = -9\sin(3x)$$

Answer:
$f''(x) = -9\sin(3x)$

Example 14.4: Product Rule Application

Find the second derivative of
$f(x) = x^2 \cdot e^x$

Solution:

Step 1: Find the first derivative using the product rule:

$$f'(x) = (x^2)' \cdot e^x + x^2 \cdot (e^x)'$$

$$f'(x) = 2x \cdot e^x + x^2 \cdot e^x = e^x(2x + x^2)$$

Step 2: Find the second derivative using the product rule again:

$$f''(x) = (e^x)' \cdot (2x + x^2) + e^x \cdot (2x + x^2)'$$

$$f'(x) = e^x \cdot (2x + x^2) + e^x \cdot (2 + 2x)$$

$$f'(x) = e^x(2x + x^2 + 2 + 2x)$$

$$f'(x) = e^x(x^2 + 4x + 2)$$

Answer:
$$f'(x) = e^x(x^2 + 4x + 2)$$

14.4 Physical Interpretation: Acceleration (Ý nghĩa Vật lý: Gia tốc)

Position, Velocity, and Acceleration

In physics, if an object moves along a straight line with position function $s(t)$ at time t:

> **Key Relationships:**
>
> - **Position:** $s(t)$ (measured in meters, feet, etc.)
> - **Velocity:** $v(t) = s'(t)$ (rate of change of position)
> - **Acceleration:** $a(t) = v'(t) = s''(t)$ (rate of change of velocity)

Interpretation:

- **Velocity** tells us how fast the position is changing
- **Acceleration** tells us how fast the velocity is changing

Sign of acceleration:

- If $a(t) > 0$: the object is **speeding up** (in the positive direction) or **slowing down** (in the negative direction)
- If $a(t) < 0$: the object is **slowing down** (in the positive direction) or **speeding up** (in the negative direction)
- If $a(t) = 0$: the velocity is **constant** at that instant

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Example 14.5: Motion Problem

A particle moves along a line with position function $s(t) = t^3 - 6t^2 + 9t + 1$ **(meters), where** t **is measured in seconds.**

(a) Find the velocity and acceleration functions.

(b) Find the velocity and acceleration at $t = 2$ **seconds.**

(c) When is the particle at rest?

Solution:

(a) Finding velocity and acceleration:

Velocity:

$$v(t) = s'(t) = 3t^2 - 12t + 9$$

Acceleration:

$$a(t) = v'(t) = s''(t) = 6t - 12$$

(b) At $t = 2$

seconds:

Velocity:

$v(2) = 3(2)^2 - 12(2) + 9 = 12 - 24 + 9 = -3$
m/s

Acceleration:

$a(2) = 6(2) - 12 = 12 - 12 = 0$
m/s²

(c) The particle is at rest when
$v(t) = 0$
:

$$3t^2 - 12t + 9 = 0$$

$$t^2 - 4t + 3 = 0$$

$$(t - 1)(t - 3) = 0$$

$t = 1$
or
$t = 3$

Answer:

- (a)
  $v(t) = 3t^2 - 12t + 9$
  ,
  $a(t) = 6t - 12$
- (b) At
  $t = 2$
  : velocity = -3 m/s, acceleration = 0 m/s²
- (c) The particle is at rest at
  $t = 1$
  second and
  $t = 3$
  seconds

Example 14.6: Free Fall

An object is dropped from a height of 100 meters. Its height above the ground at time t seconds is given by $h(t) = 100 - 4.9t^2$ (meters).

(a) Find the velocity and acceleration functions.

(b) What is the acceleration at any time?

(c) When does the object hit the ground, and what is its velocity at that moment?

Solution:

(a) Velocity and acceleration:

Velocity: $v(t) = h'(t) = -9.8t$ m/s

Acceleration: $a(t) = v'(t) = h''(t) = -9.8$ m/s²

(b) Acceleration at any time:

The acceleration is constant: $a(t) = -9.8$ m/s² (gravitational acceleration)

(c) When does it hit the ground?

The object hits the ground when $h(t) = 0$:

$$100 - 4.9t^2 = 0$$

$$t^2 = \frac{100}{4.9} \approx 20.41$$

$t \approx 4.52$ seconds

Velocity at impact:

$v(4.52) = -9.8(4.52) \approx -44.3$ m/s

Answer:

- (a) $v(t) = -9.8t$, $a(t) = -9.8$ m/s²
- (b) Constant acceleration of -9.8 m/s² (gravity)
- (c) Hits ground at $t \approx 4.52$ seconds with velocity ≈ -44.3 m/s

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14.5 Concavity and the Second Derivative (Tính Lõm và Đạo hàm Cấp hai)

The second derivative provides information about the **concavity** (curvature) of a function's graph.

> **Concavity Test:**
>
> Let f be a function with second derivative $f''(x)$ on an interval I.
>
> - If $f''(x) > 0$ for all x in I

> , then f is **concave up** on I
>
> - If $f''(x) < 0$ for all x in I, then f is **concave down** on I

Visual Interpretation:

- **Concave up** ($f'' > 0$): The graph curves upward like a cup ∪ (holds water)
- **Concave down** ($f'' < 0$): The graph curves downward like a cap ∩ (spills water)

Inflection Points

> **Definition: Inflection Point**
>
> A point $(c, f(c))$ on the graph of f is an **inflection point** if the concavity changes at $x = c$.
>
> **Finding inflection points:**

> 1. Find all values where
> $f''(x) = 0$
>   or
> $f''(x)$
>   does not exist
> 2. Check if the sign of
> $f''(x)$
>   changes at these points

Example 14.7: Analyzing Concavity

For the function
$f(x) = x^3 - 3x^2 + 2$
:

(a) Find the second derivative.

(b) Determine where the function is concave up and concave down.

(c) Find any inflection points.

Solution:

(a) Finding the second derivative:

$$f'(x) = 3x^2 - 6x$$

$$f''(x) = 6x - 6 = 6(x - 1)$$

(b) Analyzing concavity:

Set
$f''(x) = 0$
:
$6(x - 1) = 0$
$x = 1$

Test intervals:

Interval	Test point	$f''(x)$	Concavity
$x < 1$	$x = 0$	$f''(0) = -6 < 0$	Concave down
$x > 1$	$x = 2$	$f''(2) = 6 > 0$	Concave up

(c) Inflection point:

Since the concavity changes at

$x = 1$

, there is an inflection point at:

$$f(1) = (1)^3 - 3(1)^2 + 2 = 1 - 3 + 2 = 0$$

Inflection point:
$(1, 0)$

Answer:

- (a)
  $f''(x) = 6x - 6$

- (b) Concave down on
  $(-\infty, 1)$
  ; concave up on
  $(1, \infty)$

- (c) Inflection point at
  $(1, 0)$

Example 14.8: Exponential Function Concavity

Determine the concavity of
$f(x) = e^{-x^2}$

Solution:

Step 1: Find the first derivative using the chain rule:

$$f'(x) = e^{-x^2} \cdot (-2x) = -2xe^{-x^2}$$

Step 2: Find the second derivative using the product rule:

$$f''(x) = (-2x)' \cdot e^{-x^2} + (-2x) \cdot (e^{-x^2})'$$

$$f''(x) = -2 \cdot e^{-x^2} + (-2x) \cdot e^{-x^2} \cdot (-2x)$$

$$f''(x) = -2e^{-x^2} + 4x^2e^{-x^2}$$

$$f''(x) = e^{-x^2}(4x^2 - 2)$$

Step 3: Find where $f''(x) = 0$:

$$e^{-x^2}(4x^2 - 2) = 0$$

Since $e^{-x^2} > 0$ for all x:

$$4x^2 - 2 = 0$$

$$x^2 = \frac{1}{2}$$

$$x = \pm\frac{1}{\sqrt{2}} = \pm\frac{\sqrt{2}}{2}$$

Step 4: Test intervals:

Interval	Test point	Sign of $f''(x)$	Concavity

$x < -\frac{\sqrt{2}}{2}$	$x = -1$	$f''(-1) = e^{-1}(4-2) > 0$	Concave up
$-\frac{\sqrt{2}}{2} < x < \frac{\sqrt{2}}{2}$	$x = 0$	$f''(0) = e^{0}(-2) < 0$	Concave down
$x > \frac{\sqrt{2}}{2}$	$x = 1$	$f''(1) = e^{-1}(4 - 2) > 0$	Concave up

Answer:

- Concave up on $(-\infty, -\frac{\sqrt{2}}{2})$ and $(\frac{\sqrt{2}}{2}, \infty)$
- Concave down on $(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$
- Inflection points at $x = \pm\frac{\sqrt{2}}{2}$

14.6 Practice Exercises (Bài tập Luyện tập)

Exercise 14.1: Find the second derivative of $f(x) = 5x^4 - 3x^3 + 7x - 2$

Exercise 14.2: Find $\frac{d^2y}{dx^2}$ if $y = \ln(x^2 + 1)$

Exercise 14.3: A particle's position is given by $s(t) = 2t^3 - 9t^2 + 12t$

(meters). Find:

- (a) The velocity and acceleration functions
- (b) When the particle is at rest
- (c) The acceleration when the particle is at rest

Exercise 14.4: Determine the intervals where
$f(x) = x^4 - 4x^3$
is concave up and concave down. Find any inflection points.

14.7 Chapter Summary (Tóm tắt Chương)

Key Concepts

1. **Second Derivative Definition:**
   - $f''(x) = \frac{d}{dx}[f'(x)]$
   - The derivative of the first derivative
2. **Physical Interpretation:**
   - Position:
     $s(t)$
   - Velocity:
     $v(t) = s'(t)$
   - Acceleration:
     $a(t) = v'(t) = s''(t)$
3. **Concavity:**
   - $f''(x) > 0$
     $\Longrightarrow$ concave up (∪)
   - $f''(x) < 0$
     $\Longrightarrow$ concave down (∩)
   - Inflection point: where concavity changes

Common Second Derivatives

Function	First Derivative	Second Derivative
x^n	nx^{n-1}	$n(n-1)x^{n-2}$
e^x	e^x	e^x
e^{kx}	ke^{kx}	$k^2 e^{kx}$
$\ln x$	$\frac{1}{x}$	$-\frac{1}{x^2}$
$\sin x$	$\cos x$	$-\sin x$
$\cos x$	$-\sin x$	$-\cos x$

Problem-Solving Strategy

For motion problems:

1. Identify the position function
   $s(t)$
2. Find velocity:
   $v(t) = s'(t)$
3. Find acceleration:
   $a(t) = s''(t)$
4. Interpret the signs and values

For concavity analysis:

1. Find
   $f''(x)$

2. Solve
   $$f''(x) = 0$$
   to find potential inflection points
3. Test intervals to determine where
   $$f'' > 0$$
   or
   $$f'' < 0$$
4. Verify that concavity changes at inflection points

14.8 Review Exercises (Bài tập Ôn tập)

Review 14.1: Find the second derivative of:

- (a)
  $$f(x) = x^5 - 2x^4 + 3x^2 - 7$$
- (b)
  $$g(x) = e^{3x} \sin x$$
- (c)
  $$h(x) = \frac{x^2}{x+1}$$

Review 14.2: The height of a ball thrown upward is
$$h(t) = -4.9t^2 + 20t + 2$$
meters.

- (a) Find the velocity and acceleration functions
- (b) What is the maximum height reached?
- (c) When does the ball hit the ground?

Review 14.3: For
$$f(x) = x^4 - 6x^2 + 5$$
:

- (a) Find all critical points (where
  $$f'(x) = 0$$
)

- (b) Determine the concavity on different intervals
- (c) Find all inflection points

Review 14.4: Prove that
$f(x) = e^x$
is concave up for all
x
.

Memory Aids (Gợi nhớ)

Remembering the relationship:

- **Position → Velocity → Acceleration**
- $s(t)$
  $\rightarrow$
  $s'(t) = v(t)$
  $\rightarrow$
  $s''(t) = a(t)$

Concavity mnemonic:

- **Positive** second derivative → **U**pward curve (∪) → holds water
- **Negative** second derivative → dow**N**ward curve (∩) → spills water

Common mistake to avoid:

- Don't confuse
  $f''(x) = 0$
  with an inflection point automatically
- You must verify that the **concavity changes** at that point

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End of Chapter 14